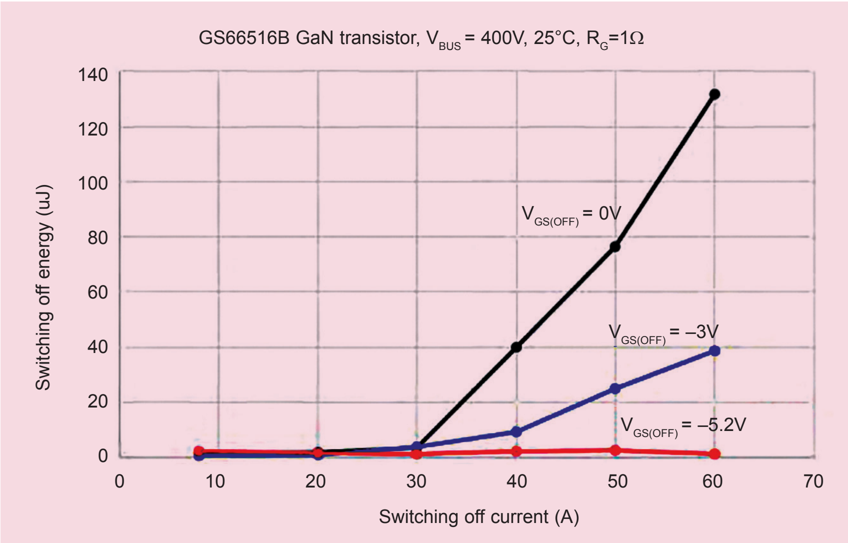

Fig. 4: Switching-off loss vs current

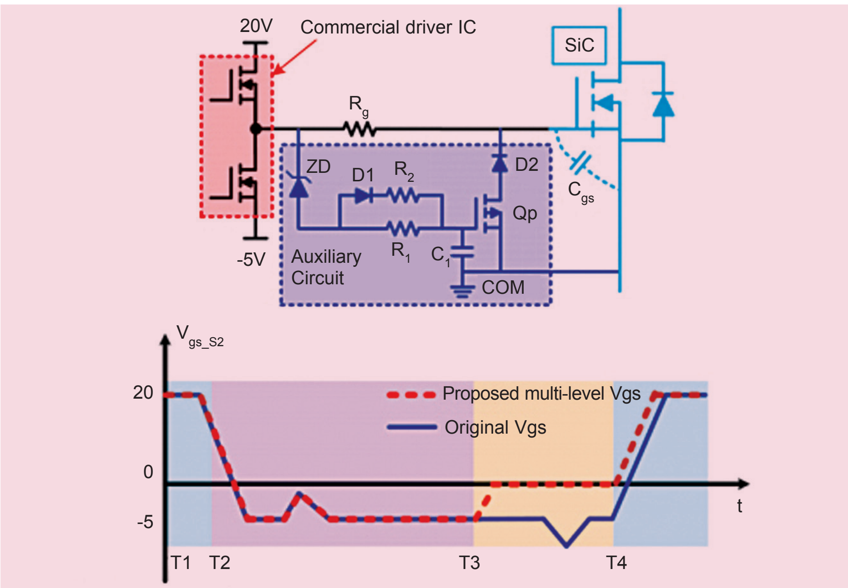

Fig. 5: Proposed SMGD circuit for crosstalk suppression and waveform

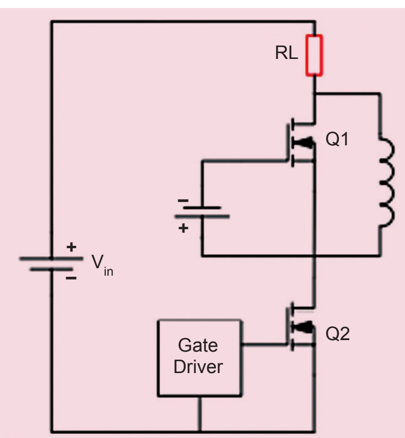

Fig. 6: Series connection of frequency-dependent resistor and inductor

external gate resistance, and internal gate resistance of the MOSFET or IGBT. It can be easily observed that the external gate resistance is the only component that tunes the gate drive waveform.

It is very important to select an optimum gate resistor for a high-performance design. Following points should be considered while selecting an external gate drive resistor:

(i) The switching characteristics of both turn-on and turn-off are dependent on value of the gate resistance. The greater the gate resistance, the longer the switching time and the greater the switching loss. Also, as the gate resistance increases, the surge voltage during switching becomes smaller. So, there is a trade-off between switching speed and overshoots.

(ii) The greater the RG, the more unlikely a dv/dt shoot-through current becomes.

(iii) Various switching characteristics are varied for stray inductance. Therefore, the gate resistance needs to be designed on the lower stray inductance condition.

All the above points suggest that the value of gate resistance must be maximised in order to suppress overshoots, and all other problems induced by high dV/dt and di/dt. But increasing the resistance will lower the switching speed. Therefore, an optimal value is crucial.

It is recommended to select the gate resistor that will give your design a quality factor Q between 0.5 and 1. A quality factor greater than 0.5 offers faster turn-on and turn-off when required.

For selecting an optimal value, record the gate drive ring with no external resistor. This is the ring frequency f_R. Plug this value in the following equation to calculate the source inductance L_S. The value of Ciss can be found in the device datasheet.

$$L_S=1/(C_{ISS}(2\pi f_R)$$

Determine the gate resistance from this equation, where R_G is the gate resistance:

$$Q = \omega L_S/R_G$$

You can also select the optimal value by simulating, starting from 0 ohms of external resistance. Fig. 3 show the impact of external gate resistance on overshoots.

The trade-off between fast rise and fall times vs overshoots and oscillations is why the external gate resistor element of the gate-drive design is so valuable.

2. Implementing negative drive voltage. Negative drive voltage is not required, but it is often beneficial. In many cases, negative drive voltage can increase noise immunity and reduce switching loss.

While driving high-power GaN systems, there is a risk of Miller effect induced turn-on when a high rate of change of voltage (high dV/dt) is seen across the power device. In such scenarios, current is injected onto